

Tamil Nadu Assembly Election Analysis

2021 vs 2026 — SQL Server + Power BI Portfolio Project

Constituency-level analysis across 234 Assembly constituencies

SAGAR TAKORE

1. Project Overview

This project analyzes Tamil Nadu Assembly election results across 234 constituencies, comparing the 2021 and 2026 elections. The analysis focuses on constituency-level seat movement, winning-party transitions, winning margins, and voter turnout changes.

SQL Server was used for data preparation, validation, winner/runner-up identification, comparative calculations, and creation of an analysis-ready SQL view. Power BI was then used to build an interactive three-page dashboard for summary analysis, party performance, and constituency-level exploration.

2. Analytical Questions

Seat Movement

- How many constituencies changed their winning party between 2021 and 2026?
- Which party-to-party winner transitions occurred most frequently?

Winning Margins

- How did winning margins change between the two elections?
- How many constituencies experienced increased versus decreased winning margins?

Turnout

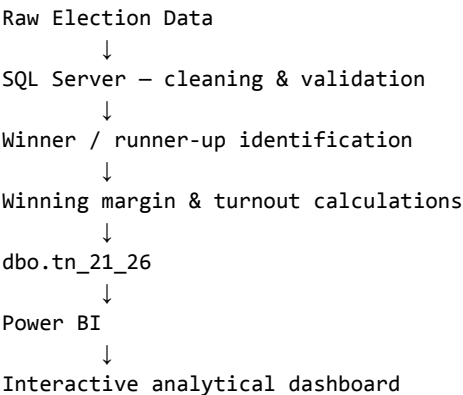
- How did voter turnout change between 2021 and 2026?
- How did turnout changes vary by region?
- Which constituencies recorded the largest turnout increases?

Constituency-Level Exploration

- How did the winner, winning party, votes, margin, and turnout change for an individual constituency?

3. Data Pipeline

The project follows a layered workflow in which detailed election data is cleaned and validated in SQL before being transformed into an analysis-ready view for Power BI.



4. Data Grain and Analytical View

The final analytical grain is one row per Assembly Constituency. The Power BI dashboard uses the SQL view `dbo.tn_21_26` as its main analysis-ready source.

Layer	Purpose
Raw / detailed result data	Source election records at candidate level
SQL cleaning and validation	Data quality checks, normalization, joins and validation
dbo.tn_21_26	One-row-per-constituency comparison of 2021 and 2026
Power BI	Interactive reporting, DAX measures, filtering and visualization

5. SQL Analysis and Transformation

Winner and Runner-up Identification

Candidate results were ranked within each Assembly Constituency using a window function. The highest vote total was treated as the winner and the second-highest as the runner-up.

```
ROW_NUMBER() OVER (
  PARTITION BY ac_number
  ORDER BY votes DESC
)
```

Winning Margin

Winning Margin = Winner Votes - Runner-up Votes

Party Transition

Party Transition = Winner Party 2021 + ' -> ' + Winner Party 2026

Seat Status

If Winner Party 2021 = Winner Party 2026 → No Change

If Winner Party 2021 <> Winner Party 2026 → Flipped

Turnout Change

Turnout Change = Turnout 2026 - Turnout 2021

6. Data Validation

The analysis was validated before being used in Power BI. The validation process checked the constituency grain, identifiers, candidate records, vote fields, winners, margins, turnout, and final analytical counts.

Validation	Expected / Result
2021 constituencies	234
2026 constituencies	234
AC number range	1–234; validated
Duplicate candidate check	Validated using (ac_number, candidate)
Vote field validation	Numeric values validated
Final analytical grain	1 row per constituency
Final analytical rows	234
Seats flipped	163
Seats with no change	71
Margin decreased	138
Margin increased	96
Turnout increased	234 of 234
Turnout decreased	0
Turnout unchanged	0

7. Final Validation SQL

The following is a compact final QA script that can be retained with the project. It is intended to validate the finished analytical view before publishing or refreshing the dashboard.

```
-- 1. Row count
SELECT COUNT(*) AS total_rows
FROM dbo.tn_21_26;

-- 2. Distinct constituency count
SELECT COUNT(DISTINCT ac_number) AS distinct_ac
FROM dbo.tn_21_26;

-- 3. Duplicate AC numbers
SELECT ac_number, COUNT(*) AS row_count
```

```

FROM dbo.tn_21_26
GROUP BY ac_number
HAVING COUNT(*) > 1;

-- 4. Missing critical fields
SELECT
    SUM(CASE WHEN ac_number IS NULL THEN 1 ELSE 0 END) AS missing_ac,
    SUM(CASE WHEN constituency IS NULL THEN 1 ELSE 0 END) AS missing_constituency,
    SUM(CASE WHEN winner_2021 IS NULL THEN 1 ELSE 0 END) AS missing_winner_2021,
    SUM(CASE WHEN winner_2026 IS NULL THEN 1 ELSE 0 END) AS missing_winner_2026,
    SUM(CASE WHEN margin_2021 IS NULL THEN 1 ELSE 0 END) AS missing_margin_2021,
    SUM(CASE WHEN margin_2026 IS NULL THEN 1 ELSE 0 END) AS missing_margin_2026,
    SUM(CASE WHEN turnout_2021 IS NULL THEN 1 ELSE 0 END) AS missing_turnout_2021,
    SUM(CASE WHEN turnout_2026 IS NULL THEN 1 ELSE 0 END) AS missing_turnout_2026
FROM dbo.tn_21_26;

-- 5. Seat status validation
SELECT seat_status, COUNT(*) AS constituencies
FROM dbo.tn_21_26
GROUP BY seat_status;

-- 6. Margin direction validation
SELECT
    CASE
        WHEN margin_change < 0 THEN 'Margin Decreased'
        WHEN margin_change > 0 THEN 'Margin Increased'
        ELSE 'No Change'
    END AS margin_direction,
    COUNT(*) AS constituencies
FROM dbo.tn_21_26
GROUP BY
    CASE
        WHEN margin_change < 0 THEN 'Margin Decreased'
        WHEN margin_change > 0 THEN 'Margin Increased'
        ELSE 'No Change'
    END;

-- 7. Turnout direction validation
SELECT
    SUM(CASE WHEN turnout_change > 0 THEN 1 ELSE 0 END) AS increased,
    SUM(CASE WHEN turnout_change < 0 THEN 1 ELSE 0 END) AS decreased,
    SUM(CASE WHEN turnout_change = 0 THEN 1 ELSE 0 END) AS unchanged
FROM dbo.tn_21_26;

```

8. Key Findings

Finding 1 — Significant Seat-Level Party Movement

The analysis classified 163 of 234 constituencies as flipped and 71 as having no change in winning party between 2021 and 2026.

Party Transition	Constituencies
DMK → TVK	65
DMK → DMK	40
AIADMK → TVK	26
AIADMK → AIADMK	22
DMK → AIADMK	22
AIADMK → DMK	15

These are descriptive transitions observed in the dataset. Transition counts alone do not establish why a constituency changed winners.

Finding 2 — Average Winning Margins Became Smaller

Election	Average Winning Margin
2021	22,870.55 votes
2026	16,784.24 votes
Change	-6,086.30 votes

At constituency level, 138 constituencies recorded a decrease in winning margin, while 96 recorded an increase. A smaller margin means a smaller vote gap between the winner and runner-up; the metric does not by itself explain the cause of that change.

Finding 3 — Turnout Increased Across All Constituencies

Election	Average Turnout
2021	73.37%
2026	85.82%
Change	+12.45 percentage points

Turnout increased in all 234 constituencies in the analyzed data: 234 increased, 0 decreased, and 0 remained unchanged.

Regional Turnout Summary

Region	2021 Avg.	2026 Avg.	Change
Chennai Metro	63.41%	84.21%	+20.80 pp
Kongu	73.06%	87.83%	+14.77 pp
North	78.05%	89.60%	+11.55 pp
South	71.27%	81.75%	+10.47 pp
Central	78.92%	89.26%	+10.35 pp
Delta	74.91%	84.05%	+9.14 pp

9. Power BI Dashboard

The final Power BI report contains three pages. Each page has a distinct analytical purpose rather than repeating the same charts.

Page	Purpose	Main Content
ELECTION ANALYSIS	Overall findings	KPIs, seat movement, margin direction, party transitions, regional turnout, top turnout changes
PARTY PERFORMANCE	Winning-party movement	Transition ranking, 2021→2026 matrix, regional winner-party distribution
CONSTITUENCY DETAILS	Row-level exploration	Search/filter controls and one-row-per-constituency comparison table

Interactive Features

- Page navigation across all three report pages
- Region, reserved-category, winner-party and seat-status slicers
- Constituency search on the detail page
- Conditional formatting for margin and turnout changes
- Constituency-level comparison of 2021 and 2026 results

10. Tools and Technologies

Technology	Use in Project
SQL Server	Data cleaning, transformation, validation, analytical calculations and view creation
T-SQL	CTEs, window functions, joins, aggregations and conditional logic
Power BI	Interactive dashboard, DAX measures, slicers and visual analytics
DAX	KPI measures and analytical calculations used in the report

11. Analytical Limitations

The project is descriptive. It identifies differences between the 2021 and 2026 election results but does not establish causal relationships.

- Higher turnout does not by itself establish which party benefited from the additional participation.
- A smaller winning margin does not explain why the vote gap changed.
- Party transition counts describe winner changes but are not equivalent to vote-share changes.
- Causal or explanatory analysis would require additional variables such as demographic, socioeconomic, historical, campaign, or other contextual data.

12. Portfolio Presentation Structure

For a Data Analyst portfolio, the project can be presented in the following sequence:

- Problem statement / analytical objective
- Data sources and data grain
- SQL data preparation and validation
- Winner, runner-up, margin and turnout calculations
- Power BI data model and DAX
- Three dashboard pages
- Three key findings
- Analytical limitations

13. Project Summary

This project demonstrates an end-to-end analytical workflow: transforming detailed election data in SQL Server, validating the resulting dataset, creating an analysis-ready constituency-level view, and building an interactive Power BI report. The final dashboard separates overall findings, party performance, and constituency-level exploration while keeping the underlying calculations traceable to the SQL analysis.